# Exercise Sheet 14

Throughout, $K$ is a compact Lie group, $\mathscr{R}(K)$ is its algebra of representative functions, and $\Delta, \varepsilon, S, *$ are as in Lecture 14. For an object $V$ of $\mathrm{Rep}_{\mathrm{fd}}(K)$ with basis $(e_j)$ and dual basis $(e^i)$ we write $D^V_{ij}(g) = e^i(\Pi_V(g)e_j)$.

**Exercise 14.1.** Let $f \in C(K)$. For $h \in K$ define $(R_h f)(g) = f(gh)$, and let $W_f = \mathrm{span}\{R_h f : h \in K\}$.

(a) Show that $R_h R_k = R_{hk}$.
(b) Show that $W_f$ is finite-dimensional if $f \in \mathscr{R}(K)$.
(c) Suppose $W = W_f$ has finite dimension $n$. Show that there are points $x_1, \dots, x_n \in K$ and a basis $\varphi_1, \dots, \varphi_n$ of $W$ with $\varphi_j(x_i) = \delta_{ij}$.
(d) In the situation of (c), show that $\rho(h) = R_h|_W$ is a continuous representation of $K$ on $W$, and that $f = c^W_{\epsilon,f}$, where $\epsilon \in W^*$ is evaluation at $e$, given by $\epsilon(\varphi) = \varphi(e)$. Conclude that $f \in \mathscr{R}(K)$.
(e) Assume that the identity component $K_0$ is open in $K$. Show that the indicator function $1_{K_0}$ lies in $\mathscr{R}(K)$.

**Exercise 14.2.** Let $G$ be a finite group with the discrete topology. For $x \in G$ let $\delta_x$ be the function equal to 1 at $x$ and 0 elsewhere.

(a) Show that $\mathscr{R}(G)$ is the space of all functions on $G$.
(b) Show that

$$\Delta\delta_x = \sum_{y \in G} \delta_y \otimes \delta_{y^{-1}x}, \qquad \varepsilon\delta_x = \delta_{x,e}, \qquad S\delta_x = \delta_{x^{-1}}, \qquad \delta_x^* = \delta_x.$$

(c) Verify $m(S \otimes \mathrm{id})\Delta\delta_x = \varepsilon(\delta_x)1$ directly.
(d) Show that every character of $\mathscr{R}(G)$ is $\mathrm{ev}_g$ for some $g$. What is $\mathcal{X}(G)$?
(e) Explain why $\mathrm{Rep}_{\mathrm{fd}}(G)$ is a fusion category. Contrast this with the result of Lecture 14 that an infinite compact Lie group has infinitely many irreducible representations.

**Exercise 14.3.** Let $a, b, c, d$ be the coefficient functions of $\mathrm{SU}(2)$ from Lecture 14, so

$$g = \begin{pmatrix} a(g) & b(g) \\ c(g) & d(g) \end{pmatrix}.$$

Lecture 14 shows that every element of $\mathscr{R}(\mathrm{SU}(2))$ is $p(a, b, c, d)$ for a polynomial $p$ in four variables.

(a) Using the formulas for $\Delta$ and $S$, verify $m(S \otimes \mathrm{id})\Delta a = 1$, $m(S \otimes \mathrm{id})\Delta b = 0$ and $m(\mathrm{id} \otimes S)\Delta a = 1$.
(b) Let $A \in \mathrm{SL}(2, \mathbb{C})$. You may use the polar decomposition $A = U \exp(H)$ with $U \in \mathrm{SU}(2)$ and $H$ Hermitian of trace 0. Show that if $p(a, b, c, d) = 0$ on $\mathrm{SU}(2)$, then $p(A_{11}, A_{12}, A_{21}, A_{22}) = 0$.
(c) Deduce that the formula $\psi_A(p(a, b, c, d)) = p(A)$ gives a well-defined character $\psi_A$, and that $A \mapsto \psi_A$ is a group isomorphism $\mathrm{SL}(2, \mathbb{C}) \to \mathcal{X}(\mathrm{SU}(2))$.

(d) Show that $\psi_A$ is a $*$-character exactly when $A \in \mathrm{SU}(2)$. Test this on the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

using the function $b$.

**Exercise 14.4.** Let $\theta \in \mathrm{Aut}_\otimes(\omega)$ be a tensor automorphism of the fibre functor of $\mathrm{Rep}_{\mathrm{fd}}(K)$.

(a) Show that if $c^V_{\lambda,v} = 0$, then $\lambda(\theta_V v) = 0$.
(b) Show that $\psi_\theta(c^V_{\lambda,v}) = \lambda(\theta_V v)$ gives a well-defined linear map $\psi_\theta \colon \mathscr{R}(K) \to \mathbb{C}$.
(c) Show that $\psi_\theta$ is a character.
(d) Show that $\psi_{\theta\theta'} = \psi_\theta \psi_{\theta'}$, and that $\theta \mapsto \psi_\theta$ is injective.

**Exercise 14.5.** Let $\psi$ be a character of $\mathscr{R}(K)$. For each object $V$ define $\theta^\psi_V \in \mathrm{End}(V)$ by $\lambda(\theta^\psi_V v) = \psi(c^V_{\lambda,v})$.

(a) Explain why $\theta^\psi_V$ is well defined. Show that $\theta^\psi$ is natural and monoidal, and that $\theta^\psi_{\mathbb{C}} = 1$.
(b) Show that $\theta^\psi_V \theta^{\psi'}_V = \theta^{\psi\psi'}_V$ and $\theta^\varepsilon_V = \mathrm{id}$. Deduce that $\theta^\psi \in \mathrm{Aut}_\otimes(\omega)$, and that $\theta \mapsto \psi_\theta$ from Exercise 14.4 is a group isomorphism.
(c) Let $\theta \in \mathrm{Aut}_\otimes(\omega)$, and let $V$ carry a $K$-invariant inner product with orthonormal basis $(e_j)$. Let $A$ be the matrix of $\theta_V$. Show $\psi_\theta((D^V_{ij})^*) = (A^{-1})_{ji}$. Deduce that $\theta$ is unitary exactly when $\psi_\theta$ is a $*$-character.

**Exercise 14.6.** Let $V$ be an object of $\mathrm{Rep}_{\mathrm{fd}}(K)$ with basis $(e_i)$ and dual basis $(e^i)$. Define

$$\mathrm{ev}_V(\lambda \otimes v) = \lambda(v), \qquad \mathrm{coev}_V(1) = \sum_i e_i \otimes e^i.$$

(a) Show that $\mathrm{coev}_V(1)$ is independent of the basis and $K$-invariant.
(b) Show that $\mathrm{ev}_V \colon V^* \otimes V \to \mathbb{C}$ is a morphism.
(c) Verify the two zigzag identities

$$(\mathrm{id}_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes \mathrm{id}_V) = \mathrm{id}_V,$$
$$(\mathrm{ev}_V \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_{V^*} \otimes \mathrm{coev}_V) = \mathrm{id}_{V^*}.$$

Use the standard identifications of $(X \otimes Y) \otimes Z$ with $X \otimes (Y \otimes Z)$.
(d) Show that $\tau(v \otimes w) = w \otimes v$ is a morphism with $\tau^2 = \mathrm{id}$. Compute the scalar $\mathrm{ev}_V \circ \tau_{V,V^*} \circ \mathrm{coev}_V$.
(e) For $K = \mathrm{U}(1)$, show that $\mathbb{C}^*_n \cong \mathbb{C}_{-n}$, and relate this to $Sz^n = z^{-n}$.

**Exercise 14.7.** Let $a, b, c, d$ be the coefficient functions of SU(2) from Lecture 14. On $\mathbb{C}^2$ with standard basis $(e_1, e_2)$ and the standard inner product,

$$ge_1 = a(g)e_1 + c(g)e_2, \qquad ge_2 = b(g)e_1 + d(g)e_2.$$

Let $V_2 = \mathrm{Sym}^2(\mathbb{C}^2)$, of dimension 3 and spin 1, with $g(uv) = (gu)(gv)$ for $u, v \in \mathbb{C}^2$.

(a) Compute $g(e_1^2)$, $g(e_1e_2)$ and $g(e_2^2)$.

(b) Let $\iota\colon V_2 \to \mathbb{C}^2 \otimes \mathbb{C}^2$ be the linear map with $\iota(uv) = \frac{1}{2}(u \otimes v + v \otimes u)$. Explain why $\iota$ is an equivariant injection. Give $V_2$ the pulled-back inner product $\langle x, y\rangle = \langle \iota x, \iota y\rangle$. Show that it is SU(2)-invariant and that

$$f_1 = e_1^2, \qquad f_2 = \sqrt{2}\, e_1e_2, \qquad f_3 = e_2^2$$

is orthonormal. Compute the matrix $D(g)$ in the basis $(f_1, f_2, f_3)$.

(c) Using only that $\Delta$ is an algebra homomorphism and the formulas for $\Delta a, \dots, \Delta d$, verify $\Delta D_{22} = \sum_k D_{2k} \otimes D_{k2}$.

(d) Verify $SD_{12} = (D_{21})^*$. Then show that in the basis $(e_1^2, e_1e_2, e_2^2)$ the corresponding entries $D'_{12}, D'_{21}$ satisfy $SD'_{12} \neq (D'_{21})^*$.

(e) Verify $\sum_k (D_{k1})^* D_{k1} = 1$ as functions on SU(2).